

Proof. **Trivial cases.**

- **If $a \leq 0$.** Since $X \geq 0$ a.s., the event $\{X \geq a\}$ is the whole space Ω , hence

$$\mathbb{P}(X \geq a) = 1.$$

Because $a \leq 0$ the right-hand side $\frac{\mathbb{E}[X]}{a}$ is non-negative (indeed it is $+\infty$ or undefined), so the inequality holds vacuously. From now on we restrict to $a > 0$.

- **If $\mathbb{E}[X] = 0$.** For a non-negative random variable, $\mathbb{E}[X] = 0$ forces $X = 0$ $\mathbb{P}$ -a.s. Consequently $\mathbb{P}(X \geq a) = 0$ for any $a > 0$, and the inequality becomes $0 \leq 0$, i.e. it holds with equality.

Thus we may assume

$$a > 0, \quad \mathbb{E}[X] > 0.$$

Indicator of the event $\{X \geq a\}$. Define the indicator (characteristic) function

$$I(\omega) := \mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & \text{if } X(\omega) \geq a, \\ 0, & \text{if } X(\omega) < a. \end{cases}$$

The indicator is measurable, takes values only in $\{0, 1\}$, and by the *Expectation of an indicator* theorem,

$$\mathbb{E}[I] = \mathbb{P}(X \geq a). \tag{1}$$

Pointwise comparison. For every $\omega \in \Omega$ we have

$$X(\omega) \geq a I(\omega). \tag{2}$$

Indeed,

- If $X(\omega) \geq a$, then $I(\omega) = 1$ and (2) reads $X(\omega) \geq a$, true.
- If $X(\omega) < a$, then $I(\omega) = 0$ and (2) reads $X(\omega) \geq 0$, true because X is non-negative.

Thus (2) holds pointwise.

Take expectations (monotonicity of the integral). The expectation (Lebesgue integral) is monotone: from (2) we obtain

$$\mathbb{E}[X] \geq \mathbb{E}[a I].$$

Since a is a deterministic constant, linearity gives

$$\mathbb{E}[a I] = a \mathbb{E}[I]. \tag{3}$$

Combining the two displays,

$$\mathbb{E}[X] \geq a \mathbb{E}[I]. \tag{4}$$

Replace $\mathbb{E}[I]$ by the probability. Substituting (??) into (??) yields

$$\mathbb{E}[X] \geq a \mathbb{P}(X \geq a).$$

Dividing by the positive constant a we obtain the desired inequality

$$\boxed{\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}}.$$

Equality case. Equality in the chain

$$X(\omega) \geq a I(\omega) \implies \mathbb{E}[X] = a \mathbb{E}[I]$$

requires equality in the pointwise inequality (??) for $\mathbb{P}$ -almost every ω . Hence, $\mathbb{P}$ -a.s.,

$$X(\omega) = a I(\omega).$$

Because $I(\omega) \in \{0, 1\}$, this means that X takes only the two values 0 and a (the degenerate sub-case $X \equiv 0$ corresponds to $\mathbb{P}(X \geq a) = 0$). In this situation

$$\mathbb{E}[X] = a \mathbb{P}(X \geq a),$$

so the bound is attained.

Conclusion. Under the sole hypotheses $X \geq 0$ a.s. and $a > 0$, Markov's inequality

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

holds, with equality if and only if X is almost surely supported on $\{0, a\}$ (including the trivial case $X \equiv 0$). □ □